StochasticVolatility: Heston, Bates and SABR in Julia

Simulation, closed forms, the Hagan formula, Longstaff-Schwartz and finite differences (MCS-IT) for European and American prices

In Pollis	Written to power the *Continuous-Time Stochastic Volatility Models* panel of the Finance Toolbox. The panel does not call it yet.
Missing in Julia	American prices under Bates and SABR. Of the packages read: DiffEqFinancial 2.11.0 has a Heston SDE and moments, no prices; Hedgehog 0.1.0 is Heston only, with no dividend yield (read from its source, not run); ArbitrageFreeSABR 1.0.0 has the arbitrage-free SABR PDE, not the 2002 formula; QuantLib.jl 0.1.1 does not install on Julia 1.13, and its `BatesEngine` at commit fe98714 has `expm1(nu * delta2)` where the C++ has `exp(nu + delta2) - 1`.

Package	`StochasticVolatility.jl` 0.1.0, MIT:
	`Pkg.Registry.add(url = "https://github.com/Trumpingtons/PollisRegistry.git")`
	`Pkg.add("StochasticVolatility")`

Tests	`test/runtests.jl` runs on Julia 1.10 and the latest 1.x at every push. Of 39 published American puts under Heston, 36 to every printed digit. Of 5 American calls under Bates, 3 agree and 2 are open. Of 14 American options under SABR, all within .00097 of QuantLib.

Why a package

The panel's procedures are hand-written loops, with no American price. American options have no closed form, so one engine alone cannot show that it is right. The package has two: least squares on simulated paths (Longstaff and Schwartz, 2001) and finite differences on the pricing equation.

What it computes

`Heston`: $dS = \mu S\,dt + \sqrt{v}\,S\,dW_1$, $dv = \kappa(\theta - v)\,dt + \xi\sqrt{v}\,dW_2$, $d\langle W_1, W_2\rangle = \rho\,dt$ (Heston, 1993). `Bates` adds compensated log-normal jumps of intensity λ, with log size of mean μ_J and deviation δ_J (Bates, 1996). `SABR`: $dF = \sigma F^{\beta}\,dW_1$, $d\sigma = \nu\sigma\,dW_2$, $\sigma_0 = \alpha$ (Hagan et al., 2002). The drift μ is the rate less the dividend yield.

Simulation

`simulate` and `simulate_terminal` with `QuadraticExponential()` (Andersen, 2008), `FullTruncation()` (Lord et al., 2010) or `Euler()`, on `PseudoRandom` or `QuasiRandom` numbers: scrambled Sobol with a Brownian bridge, and for Bates the jump bridge of Baldeaux and Roberts (2012).

European prices

`european_price`, the closed forms of Heston and Bates; `hagan_implied_vol` and `hagan_price`, the formula of Hagan et al., an approximation; `monte_carlo_price`, a vector of strikes from one simulation.

American prices

`longstaff_schwartz_price` exercises at the simulation's steps, so it is Bermudan with as many dates. By finite differences, `finite_difference_price` is the MCS-IT scheme of Haentjens and in 't Hout (2015): modified Craig-Sneyd ADI with $\theta = 1/3$ and the Ikonen-Toivanen step (Ikonen and Toivanen, 2009) for early exercise, on nonuniform meshes with first-order upwind convection. For Bates the jump integral is explicit, exact on each mesh interval under the log-normal jump. For SABR it works in F and $\log\sigma$ with central differences, and `american = false` gives the European price. The default grid is 300 prices, 150 variances (or volatilities) and 75 steps.

A worked example

```
using StochasticVolatility
model = Heston(S0 = 10, v0 = 0.0625, mu = 0.1, kappa = 5, theta = 0.16, xi = 0.9, rho = 0.1)
finite_difference_price(model, 10, 0.25; rate = 0.1, call = false)   # 0.5204, American
european_price(model, 10, 0.25; rate = 0.1, call = false)            # 0.5015, European
sabr = SABR(100.0, 2.0, 0.5, -0.7, 0.8)                              # F0, alpha, beta, rho, nu
finite_difference_price(sabr, 100, 0.25; rate = 0.05, call = false)  # 3.9466, American
hagan_price(sabr, 100, 0.25; rate = 0.05, call = false)              # 3.9388, European
```

What the tests show: American puts under Heston, against the published MCS-IT

From Haentjens and in 't Hout (2015), $K = 10$, $T = .25$ in Tables 2 and 3 and $K = 100$, $T = .25$ in Table 4. The rows show the finest grid of each table; the tests compare all 39 values, on the same method and grids.

Source and grid	S_0	8	9	10	11	12	
Table 2, $v_0 = .0625$	published	2.0000	1.1081	.5204	.2143	.0827	
$300 \times 150 \times 75$	Julia	2.0000	1.1081	.5204	.2143	.0827	✓
Table 3, $v_0 = .25$	published	2.0788	1.3339	.7962	.4486	.2433	
$300 \times 150 \times 75$	Julia	2.0788	1.3339	.7962	.4486	.2433	✓
	S_0	90	100	110			
Table 4, $v_0 = .0348$	published	10.0039	3.2126	.9305			
$300 \times 150 \times 60$	Julia	10.0039	3.2126	.9305			✓

Where they differ. 36 of 39 agree to every printed digit, the 3 others at the exercise boundary, with few steps or points. $S_0 = 8$ at $100 \times 50 \times 25$: published 2.0001, Julia 2.0000, off by 1 in the fourth decimal. $S_0 = 90$ at 20 steps: published 9.9984, Julia 10.0000, off by 16; the published value is below the exercise value 10. $S_0 = 90$ at 40 steps: published 10.0015, Julia 10.0020, off by 5. Of the 33 values not within a cent of the exercise value, the furthest is 4.9×10^{-5} away, under half of the last digit printed.

American calls under Bates: 2 of 5 open

$K = 100$, $T = .5$, rate .03, dividend yield .05, $v_0 = \theta = .04$, $\kappa = 2$, $\xi = .25$, $\rho = -.5$, $\lambda = .2$, $\mu_J = -.58$, $\delta_J = .4$. Published: Toivanen (2010) on a grid of $4096 \times 2048 \times 514$, read second-hand from Salmi, Toivanen and von Sydow (2013). The last column is the value extrapolated from the two grids of Julia, $2V_{1200} - V_{600}$, minus the published one. The 1200 grid, QuantLib 1.44 and Longstaff-Schwartz are stored in `benchmark/`, not recomputed here.

S_0	Published	Julia $600 \times 300 \times 150$	Julia $1200 \times 600 \times 300$	QuantLib $100 \times 100 \times 50$	LS, 98304 paths, 126 steps (SE)	Extrap. – publ.	
80	.328526	.332788	.330687	.331071	.33142 (.00174)	+.0001	✓
90	2.109397	2.115783	2.112723	2.108276	2.09304 (.00330)	+.0003	✓
100	6.711622	6.715916	6.713882	6.714004	6.69703 (.00723)	+.0002	✓
110	13.749337	13.749886	13.748658	13.759956	13.70704 (.00850)	–.0019	✗
120	22.143307	22.138758	22.137875	22.153021	22.08472 (.01164)	–.0063	✗

American options under SABR, against QuantLib

$F_0 = 100$, $\alpha = 2$, $\beta = .5$, $\rho = -.7$, $\nu = .8$, $T = .25$, rate .05. QuantLib 1.44 `FdSabrVanillaEngine` and Julia at $400 \times 50 \times 64$, the largest gap .00097 at call $K = 75$. LS (stored, with SE): 131072 quasi-random paths, 64 steps.

K	Put, QuantLib	Put, Julia	Put, LS	Call, QuantLib	Call, Julia	Call, LS
50	.00244	.00251	.00271 (.00029)	50	50	50
75	.12580	.12598	.12513 (.00068)	25.02978	25.03075	25.03815 (.00490)
95	2.15536	2.15524	2.14291 (.00275)	7.11113	7.11123	7.10076 (.00475)
100	3.94654	3.94666	3.93968 (.00368)	3.94646	3.94650	3.94508 (.00264)
105	6.72499	6.72566	6.71495 (.00508)	1.76876	1.76890	1.77113 (.00222)
125	25	25	25	.00276	.00217	.00338 (.00015)
200	100	100	100	0	0	0

Cases with a known answer

An American call with no dividend is never exercised early, so it must tend to the closed form. The gap below halves as the grid doubles: first order, largest for S_0 below K. $S_0 = 100$, $v_0 = \theta = .04$, rate .05, $T = .5$, $\kappa = 2$, $\xi = .3$, $\rho = -.7$; Bates with $\lambda = .2$, $\mu_J = -.58$, $\delta_J = .4$ (`benchmark/finite_difference_no_dividend_call.txt`).

Model	K	Closed form	Gap at $300 \times 150 \times 75$	at $600 \times 300 \times 150$	at $1200 \times 600 \times 300$
Heston	80	22.436161	.001693	.000861	.000434
Heston	100	6.837200	.006962	.003546	.001789
Heston	120	.504430	.013094	.006700	.003388
Bates	80	24.325572	.003762	.001886	
Bates	100	8.960508	.016100	.008099	
Bates	120	1.147607	.042395	.021430	

Under SABR at the 400 × 50 × 64 grid, American minus European at rate 0 is 3.8×10^{-9}, 1.2×10^{-6} and 6.7×10^{-5} at $K = 75$, 100 and 125, and put-call parity holds to about 10^{-13}. At $\nu = .001$, $K = 100$, Julia gives 3.939869 with $\beta = 0$ against Bachelier's 3.939865, and 3.938194 with $\beta = 1$ against Black's 3.938224. Longstaff-Schwartz under Heston (131072 quasi-random paths) is within .005 of the published values at 64 dates, and within .01 of the published Bermudan values at 60.

European prices and simulation

Function	Rows	Against	Largest difference	
`european_price`, Heston	189	QuantLib 1.44 / NMOF 2.12.0	2.8×10^{-12} / 9.0×10^{-7}	✔
`european_price`, Bates	540	QuantLib, 324 rows with $\xi < 1$ / NMOF	1.4×10^{-8} / 1.1×10^{-6}	✔
`hagan_implied_vol`, `hagan_price`	756	QuantLib 1.44	1.5×10^{-14}, relative	✔

QuantLib's `BatesEngine` has not converged at $\xi \geq 1$, so there only NMOF is used. On one file of Sobol numbers, 16 replicates of 2^{13} paths and 64 steps, the paths agree with QuantLib to 9.1×10^{-12} or closer, at 4.5 against 27.0 µs per path for Heston and 6.0 to 6.2 against 30.2 for Bates. QuantLib has no SABR process, so its side is written by hand on its generator: 4.5 to 4.6 against 4.0 µs per path, with Euler. `QuasiRandom` narrows the spread of replicate means 12 to 21 times for Heston and SABR calls, and 2.8 to 5.0 times for Bates calls.

One call price, in µs: Julia 155.1 (Heston) and 176.8 (Bates), `rtol` 10^{-12}; QuantLib C++ with Laguerre order 192, 1907.0 and 1947.4, and adaptive 10^{-13}, 678.8 for Heston; NMOF 1460.3 and 1763.0. `hagan_price`: .164 against .104. The 14 SABR options on one thread: .66 to .73 s against 3.05 s. These are not times at equal accuracy.

Still open

- Bates at $S_0 = 110$ and 120: the extrapolated values are .0019 and .0063 below Toivanen's, and the gap at 120 grows with the grid. The source is not known, and the chapter itself has not been read.
- Finite differences converge at first order on these calls: 3.7% of the price on Bates $K = 120$ at the default grid. Where a cent matters with S_0 below K, use a finer grid and compare two.
- Longstaff-Schwartz is biased, source unknown. Under Bates it is up to .0586 low, about 5 SE; under SABR about a cent low near the money. At rate 0 under SABR its American minus European runs from $-.019$ to $+.038$, largest about 4 SE, so it is not yet a clean check there; do not take it alone as one. The Euler scheme is itself biased under SABR.
- SABR call $K = 125$ is unsettled: .00276 QuantLib, .00217 Julia at 50 volatility points, .00316 at 150, .00338 LS.
- The critical spot of Table 4 is not settled: 89.68 to 89.96 by grid in finite differences, 90.2 to 90.3 in Longstaff-Schwartz.
- Andersen (2008) and Hagan et al. (2002) print no prices, so those two are checked against libraries only. Strikes very close to the forward are not yet tested in the Hagan formula.
- Calibration is not written. Among the volatility measures also in the package, the attribution of `ams_noise_test` and the constants of `meilijson` are unverified, and the range-based measures duplicate `RangeVol.jl`.

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Package written and tested with Claude Code (Anthropic) in Pollis.

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contact@pollis.org